

Kaustubh Khulbe

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EDUCATION

University of Illinois Urbana-Champaign

August 2023 - May 2026

Bachelor of Science, Computer Science

GPA 4.00/4.00

- **Relevant Coursework:** Advanced Operating Systems, Advanced Comp Arch. - Mobile and Edge Computing, Operating Systems, Applied Parallel Programming with CUDA, Comp Arch., Algorithms, Data Structures
- **Awards:** Illinois Engineering Achievement Scholar, Dean's List

PROFESSIONAL EXPERIENCE

Incoming Fall 2025 @ NVIDIA

August 2025 - December 2025

Systems Software Engineer Intern - CUDA Performance

San Jose, CA

Incoming Summer 2025 @ Amazon Annapurna Labs

May 2025 - August 2025

Compiler Engineer Intern - Graviton Compiler

Austin, TX

UIUC ADAPT Lab

August 2024 - Present

Undergraduate Researcher

Urbana, IL

- Designed a compiler toolkit to generate compiler backends for tensor ISAs and achieved fast compilation times
- Built scripts to generate XLA from graphs, and lower from XLA to AVX/Gemmini to benchmark compilation
- Fuzzing Amazon NKI compiler by writing several gold kernels to measure Tranium chip performance

SLICE Lab @ Berkeley Architecture Research

January 2025 - Present

Undergraduate Researcher

Remote

- Collaborating with Apple to realize Speculative Taint Tracking, a security mechanism, on an out-of-order CPU
- Conducting critical path and area benchmarks on rename stage
- Writing test benches in Python and SystemVerilog to validate design

University of Illinois at Urbana-Champaign

December 2024 - Present

ECE 391 (Operating Systems) Course Assistant

Urbana, IL

- Built a robust auto-grader to allow automatic interfacing with QEMU and testing student software
- Hosting office hours and midterm review sessions for a class size of 200+ students
- Expanding final OS project by integrating user-space threads, shared memory, and a pre-emptive priority scheduler

Omni RPA Inc.

May 2024 - August 2024

Software Engineer Intern

San Jose, CA

- Built a network transfer cost optimizer for an orchestrator using Go and Nebula, saving thousands of dollars
- Designed large-scale graph models to represent pricing and performance models in Snowflake and Nebula DB
- Constructed scalable microservices to automate downloading, parsing, and uploading cloud provider data

PROJECTS

QuickSend: SoC Hardware Compression | *SystemVerilog, Python*

January 2024 - Present

- Reverse-engineered software-transparent Intel iGPU compression mechanism in RTL based on the GPU.ZIP paper
- Building custom compression algorithm to improve Intel compression compressibility
- Designed testing framework for texture compression algorithms to measure compressibility under workloads

Illinux: Operating System | *C, Unix, RISC-V, QEMU*

July 2024 - December 2024

- RISC-V Unix-like OS with paging, filesystem, devices, system calls, and a pre-emptive round robin scheduling
- Developed a virtio block driver, cryptography driver, and entropy driver
- Kernel supports pipes between processes and argument passing according to POSIX standards
- **2nd place** in OS design competition in ECE department with 100+ teams

CUDA-based LeNet CNN Network | *C, C++, CUDA, nsys, ncú*

August 2024 - December 2024

- Optimized forward pass of LeNet CNN Network with streams, tensor cores, etc.
- Achieved lossless speedups of 70.45% from base trivial GPU implementation
- Used cuBLAS, half precision floating points, and constant memory on A100 GPUs to improve cache and SM usage

SKILLS

Programming Languages: C, C++, Rust, Python, CUDA, Triton, HLO

Tools and Technologies: Linux/Unix, XLA, QEMU, gdb, nsys/ncú